

CAGI → NSS 2026: Remaining Work for the Authors

Internal guideline, not for submission

Prepared 27 September 2026 · Deadline 30 September 2026, 23:59 AoE (1 October, 18:59 in Vietnam)

0. Status after review, /improve and /ready

- **Reviews.** `review_paper_report.pdf` (Reviewer A, normal mode): Weak Reject, 56/100, confidence 4/5. `report.tex` (Reviewer B, Q2 standard): Weak Reject, 58/100 as stated, which is 56/100 once B’s own 5-point category cap is applied. The two reviews agree on every overlapping point; the consolidated list is in Sect. 11 of Reviewer A’s report.
- **Done without new data** (details in `revision_notes.md`):
 - anonymized the paper;
 - fit it to 15 pages including references;
 - removed every synthetic or illustrative number;
 - corrected the “same learner” wording and added baseline B3’;
 - framed relative checkpoints as retrospective and moved the post-hoc subgroup out of the abstract and conclusion;
 - consolidated the limitations;
 - added and verified 2 references and cited Random Forest;
 - fixed the internal inconsistencies;
 - passed every static readiness gate except placeholders.
- **What blocks submission:** 19 red `\TBD` boxes in `cagi_nss2026.pdf`. They need your result files, which are not in this folder. The macro is now fail-safe. With `\pendingfalse`, any remaining `\TBD` stops the build, so a placeholder cannot slip into the submitted PDF.
- **Rule for every task:** copy numbers only from a versioned run (commit hash + config + output CSV). Record the file name next to each value in `report_audit.md` (experiment ledger). Report whatever the run gives, even if it weakens a claim; Sect. 3 below says how to word each outcome.

1. Priorities

P1 = blocks submission (placeholder, contradiction, or possible leak). **P2** = reviewers will almost certainly ask; do it if time allows before 30 Sep, otherwise for the journal version. **P3** = after submission.

2. Tasks

ID	Pri.	Task, manuscript location, and how to compute	Acceptance check / expected outcome	Due
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A. Do first: these can invalidate other numbers

ID	Pri.	Task, manuscript location, and how to compute	Acceptance check / expected outcome	Due
T0	P1	Provenance of the headline. M1 pooled PR-AUC was 0.736 in the 19 Sep draft (from <code>idea.pdf</code>) and is 0.691 now. Identify the run (commit, config, output directory) that produced 0.691, 0.688, the E4–E7 table and Table 5. Regenerate all tables from that single directory.	One run ID listed in <code>report_audit.md</code> ; every number in Tables 3–5 and the abstract traced to a file.	28 Sep
T4	P1	Prefix-length ratio (Sect. 4.3 Temporal; Sect. 6.2 case study). Find its formula in the feature code. <i>Safe</i> if it uses only $e_1..e_k$ (e.g., k divided by a quantity already observed at step k). <i>Leaks</i> if its denominator is the final length n , the endpoint index, or the relative-checkpoint percentage.	If safe: write the formula into the <code>\TBD</code> in Sect. 4.3. If it leaks: delete the feature, rerun <i>everything</i> (Tables 3–5, E4–E7, calibration, Chibi case), and update the abstract numbers. Expected: pooled PR-AUC may fall, and the 25% checkpoint result is the most likely to change.	28 Sep
T5	P1	Chibi case vs. Table 5 (Sect. 6.2). The case says the out-of-fold probability is ≈ 0.98 at $k = 2$. Table 5 says no incident is alerted at $\theta = 0.9$. Check whether 0.98 is raw or Platt-calibrated, whether Chibi is among the 10 endpoint-confirmed incidents, and recompute both from the <i>same</i> calibrated out-of-fold file.	Consistent case and table. Delete the <code>\TBD</code> sentence in the case study. If 0.98 is raw, write “raw score” and give the calibrated value.	28 Sep
B. Fill the placeholders (no new modelling beyond B3')				
T1	P1	RQ1 paired statistics (Sect. 6.1: <code>\TBD</code> {mean}, {CI}, { p }). Use the same incident-level bootstrap as Table 4 so the two are comparable: 2,000 resamples of the 15 incidents with replacement, seed 42, statistic = PR-AUC(M1) – PR-AUC(B3) on the resampled rows. Two-sided $p = 2 \min(\Pr[\Delta^* \leq 0], \Pr[\Delta^* \geq 0])$. Report the mean and the 2.5/97.5 percentiles.	Expected: CI includes 0 (the pooled gap is only +0.003). Keep the current wording “comparable”. Do <i>not</i> use Wilcoxon; Sect. 5.3 now promises a bootstrap p .	29 Sep
T3	P1	Same-learner baseline B3' (Table 3a row; Sect. 6.1). XGBoost with M1’s exact hyperparameters, per-fold <code>scale_pos_weight</code> , seed and folds, on the B3 flat feature set. Also run the T1 bootstrap for M1–B3'; this is the clean RQ1 test.	Fill the Table 3a cell and the Sect. 6.1 <code>\TBD</code> . See Sect. 3 below for wording by outcome.	29 Sep
T2	P1	False-alert rates (Table 5, 3 cells). For each $\theta \in \{0.5, 0.7, 0.9\}$, count the negative trajectories whose calibrated out-of-fold score reaches θ at <i>any</i> checkpoint. Rate = count/612. If the unit is prefix rows instead, say so in the caption.	Monotone non-increasing in θ . Write as “ $x\%$ ($c/612$)”. The previous illustrative values (0.82/0.33/0.33%) must not be reused unless the run reproduces them.	29 Sep
T6	P1	Positive trajectories and prevalence (Sect. 5.1 <code>\TBD</code> ; Table 3b, 4 cells). Count positive trajectories (the row count implies ≥ 21). For each relative checkpoint, report the positive rate of the rows it contains.	Numbers consistent with 3,354 rows at 5.0% (≈ 168 positive rows). Add one sentence on how multi-trajectory incidents are weighted in the pooled PR-AUC.	29 Sep
T7	P1	Spearman with outlier (Sect. 6.1 <code>\TBD</code>). Recompute the per-incident M1–B3 gap vs. length with all 15 incidents; name the excluded incident and why in the repository.	If ρ drops sharply, soften “correlates” to “tends to increase”, and keep the paragraph exploratory.	29 Sep

ID	Pri.	Task, manuscript location, and how to compute	Acceptance check / expected outcome	Due
T8	P1	Peel-chain thresholds (Sect. 4.3 Motif, 2 \TBD). Copy the forwarding fraction and peel fraction from the config used by the T0 run.	Two numbers, e.g., “at least 90%” / “at most 10%”, matching the code exactly.	28 Sep
T9	P1	Chibi attributions (Sect. 6.2 \TBD). After T4, extract the top-3 TreeSHAP features for Chibi at $k = 2$ from the final model.	Replace the \TBD with the actual third feature (or all three, if the order changes).	29 Sep
C. Artifact, final gate, submission				
T10	P1	Repository contents. Sect. 7 now promises the incident registry, decoded datasets, splits, <i>per-incident results</i> , and scripts that regenerate every table. Upload all of them anonymously; scrub names, e-mails, paths and git author metadata.	The link opens in a private window, and make tables (or equivalent) reproduces Tables 3–5 bit-for-bit.	29 Sep
T11	P1	Final gate. Replace every \TBD{...} by its number; set \pendingfalse ; run latexmk -pdf -halt-on-error . The build <i>fails</i> if any TBD remains. Then run pdfinfo (≤ 15 pages, empty Author); pdftotext grep -i tbd (empty); rerun audit_manuscript.py -page-limit 15 ; view every page.	0 TBD, 0 undefined refs, 0 overfull boxes, ≤ 15 pages, anonymous. Update report_audit.md to READY .	30 Sep AM
T12	P1	Submit on EasyChair (https://easychair.org/my/conference?conf=nsssocialsec2026) by 30 Sep 12:00 Vietnam time; download the uploaded PDF and open it.	Submission ID recorded.	30 Sep
D. Strengthening (reviewers will ask)				
T13	P2	Seed robustness: rerun M1, B3, B3' with 5 seeds and report mean $\pm$ s.d. of pooled PR-AUC (one sentence in Sect. 6.1).	Expected s.d. < 0.02 ; otherwise the $+0.003$ gap is noise by construction.	30 Sep / journal
T14	P2	Nested LOIO for feature selection: inside each training split, choose with vs. without the bridge-context features by inner LOIO, then score the held-out incident. Removes the optimism now disclosed in Sect. 6.1 and 7.	Report the nested PR-AUC next to 0.691; if the gap is small, delete the optimism caveat.	journal
T15	P2	Online horizons: PR-AUC and alert rate at 1, 5, 15, 30 minutes after the seed event and at fraction of seed value moved (both available live). Replaces the retrospective 25/50/75% diagnostic as the deployment evidence.	New small table; expected lower than 0.933 but well above the 0.05 floor.	journal
T16	P2	Calibration uncertainty: incident-clustered bootstrap CI for Brier and ECE; a reliability diagram.	CI reported in Sect. 6.3.	journal
T17	P2	Negative sampling: publish tolerances for chain/period/value/activity matching, the second-pass rule for the 107 complex negatives, the reason for the 5 manual controls, and per-incident negative counts.	Deterministic script in the repository.	journal
T18	P2	Trajectory-builder validity: share of verified (nonce/message-hash) vs. amount/timing bridge links; precision on a hand-checked sample of ≥ 50 links; sensitivity to hop limit $\{4, 6, 8\}$ and retained value $\{2.5, 5, 10\}\%$.	Small table; stable PR-AUC across settings supports Sect. 4.2.	journal

ID	Pri.	Task, manuscript location, and how to compute	Acceptance check / expected outcome	Due
T19	P3	Controlled representation study: same learner on Flat, Flat+Types, Flat+Motifs, Flat+Types+Motifs; plus a small GRU/1-D CNN over typed action tokens under LOIO.	Answers “is it typing, motifs, or the learner?” (Reviewer B, M8).	journal
T20	P3	Mimicry stress test (insert j benign transfers between laundering steps) and per-bridge-family detection breakdown.	Discussion material.	journal
T21	P3	More incidents ($N \geq 30$), with the long-vs-short hypothesis (split at 40 actions) <i>pre-registered</i> before looking at the new data.	Confirms or refutes the exploratory finding.	journal

3. How to word RQ1 after T1 and T3

Outcome	Wording in abstract / Sect. 6.1 / conclusion
M1–B3′ CI includes 0 (most likely)	Keep the current text: “comparable to a flat-graph baseline”; the typing effect is a hypothesis. No change to the abstract.
M1–B3′ > 0, CI excludes 0	Add to the abstract: “and exceeds the same learner on untyped features by x (95% CI $[\cdot, \cdot]$)”. Restore a typing claim in contribution 3. Do <i>not</i> add the post-hoc subgroup back.
B3′ > M1	State it plainly in Sect. 6.1 and reframe the contribution around the protocol and calibrated alerts (already the current framing).
T4 reveals a leak	Rerun first; all three rows above then apply to the new numbers.

4. Placeholder map (19 boxes in the PDF)

Location	Placeholder	Task
Sect. 4.3 Temporal	prefix-length-ratio formula	T4
Sect. 4.3 Motif	forwarding fraction; peel fraction (2)	T8
Sect. 5.1	number of positive trajectories	T6
Sect. 6.1 ¶1	PR-AUC of B3′; mean; CI; p (4)	T3, T1
Sect. 6.1 ¶2	ρ , p with outlier included	T7
Table 3a	B3′ PR-AUC	T3
Table 3b	positive rate at 25/50/75/100% (4)	T6
Table 5	false-alert rate at $\theta = 0.5/0.7/0.9$ (3)	T2
Sect. 6.2 case study	calibrated or raw; Table 5 consistency	T5
Sect. 6.2 case study	third top attribution	T4, T9

5. Schedule

Date (Vietnam time)	Work
Sun 28 Sep	T0 provenance; T4 prefix-length ratio (if it leaks, start the full rerun immediately); T8 thresholds; T5 Chibi vs. Table 5.
Mon 29 Sep	T3 B3′; T1 and T3 bootstraps; T2 false-alert rates; T6 counts and prevalence; T7; T9; T10 repository. If time allows, T13 seeds.
Tue 30 Sep, AM	T11 final gate (fill TBDs, <code>\pendingfalse</code> , build, audit, view every page). Recompile <code>cagi_nss2026_marked.tex</code> if you want to keep an internal diff.
Tue 30 Sep, by 12:00 After submission	T12 submit. The hard deadline is 1 Oct 18:59 Vietnam time; keep the rest as buffer. T14–T21 for the extended journal version.

6. Files in this folder

- `cagi_nss2026.tex/.pdf`: clean submission copy (anonymous, 15 pp., `\pendingtrue`).
- `cagi_nss2026_marked.tex/.pdf`: changes since the 27 Sep draft in blue.
- `cagi_nss2026_before_improve.tex`: the 27 Sep draft as received (named authors; synthetic values).
- `review_paper_report.tex/.pdf`: Reviewer A plus the consolidated assessment with `report.tex`.
- `revision_notes.md`: concern-by-concern change log. `report_audit.md` and `report_audit.static.md`: readiness audit.
- `Todo-list_2026-09-19.tex`: the previous to-do list, kept for reference.